

Data collection

DEMO · data_collection/demo/ · 3 participants · 1 / 600 takes at 0 mm

Character source Sensor board Demo (mouse)

DEMO: draw with the mouse, No sensor board or robot is required; demo takes are saved separately.

Experiment

name Save

Included in every character recording.

Participant

P01 name

hand Right Left age not given v consent ☐

notes excluded ☐ New Save

Demo participant 1 (P01) saved on 2026-10-06.

	Digits 0-9	Letters A-J	height	0 mm	10 mm	50 mm	100 mm	150 mm	200 mm			
Participant	Hand	A	B	C	D	E	F	G	H	I	J	Total
Demo participant 1 (P01)	R	1	0	0	0	0	0	0	0	0	0	1 Start
Demo participant 2 (P02)	R	0	0	0	0	0	0	0	0	0	0	0 Start
Demo participant 3 (P03)	R	0	0	0	0	0	0	0	0	0	0	0 Start
All		1	0	0	0	0	0	0	0	0	0	1

Tracking error (sensor board)

magnet offset mm heights mm Start

No tracking-error runs yet.

DEMO: select 0-9 or A-J; Enter starts and saves a take.
Move the mouse to draw while recording; Enter stops the ink and saves.

Close

- Protocol: 0–9 / A–J; ten takes each (200 per participant).
- Participant names and stable IDs; experiment names accompany takes.
- Enter marks capture boundaries; CSV/PNG/JSON use frozen rows.
- Separate demo files/counts; software checks + GUI rehearsal passed.

- Week 2 begins 5 October: board repair and magnet diagnostics.
- Compare one-, two- and three-magnet configurations; document geometry and orientation.
- Prototype a reloadable pen; check writing and teleoperation heights.

- Start virtual target reaching in Week 2; log completion times and endpoint errors.
- Character pilot: 40–60 takes/person (2–3 repetitions); myself + 1–2 colleagues. Not collected.
- Next: review/resume, preprocessing and participant-separated splits.

Proposed ten-week plan

Task	Start Week	End Week
Virtual task + pilot	W2	W4
Hardware + recording	W1	W3
Pilot + protocol freeze	W3	W4
Full collection + preprocessing	W4	W6
Models + tracking analysis	W5	W7
Participant evaluation	W9	W10
Analysis + report / buffer	W10	W11

MagPilot

Explicit one-character capture

Select label

>

operator Enter: start

>

participant draws

>

operator Enter: stop/save

>

inspect.

Character collection

DEMO · mouse input

Lab meeting mouse demo

Participant: P01
Session: S01

Digits 0–9 · Letters A–J
10 takes per character

- 1 Select the character
- 2 Position the pen / cursor
- 3 Enter to start
- 4 Move the mouse to draw
- 5 Enter to stop and save

Mouse movement draws while recording is active.

Escape cancels an unfinished take.
Enter repeats the selected character.

Writing Surface

Saved letter_A 1/10 — Enter to record another

0–9 / A–J select · Enter start / stop & save · Escape cancel · q quit
Move the mouse to draw while recording; Enter again repeats the selected character

DEMO · Lab meeting mouse demo | Ready: letter_A | Enter start | Saved letter_A 1/10 — Enter to record another

Saved takes

Selected: A · 1/10
Ready · Enter starts
Total: 1/200
Enter start / stop · Escape cancel

0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F
G
H
I
J

X/Y projection, 64px, trail 250, Z mode max

Actual GUI rehearsal: synthetic A; one of ten takes saved.

- Operator uses Enter for recording and demo ink; participant only moves the mouse.
- Demo mouse input; no sensor board needed.
- Stop freezes rows before matched artifacts are saved.
- Hardware pilot and new-pen validation remain pending.